

A

Physical isolation



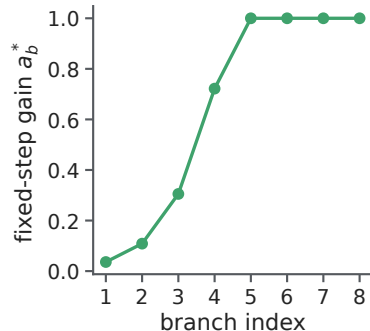
shunt $\kappa_b \geq 0$ + clamp current $\kappa_b V$

$V' = V$ but eligibility $\times G/(G + \kappa_b)$

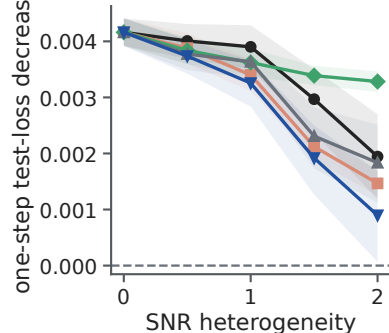
isolates physical input-resistance attenuation

B

Step-consistent profile

**C**

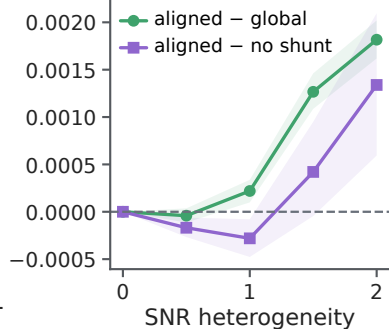
Immediate step



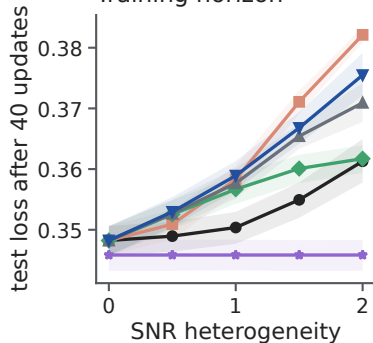
● no shunt ■ best global ◆ SNR-aligned ▲ shuffled ▼ anti-aligned ◀ exact clean BP

D

Alignment contrast

**E**

Training horizon

**F**

Final boundary

